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From Raw Data to Verifiable Trust: A Provably Safe Framework for Machine-Sensor Calibration in Industrial Cyboquatic Systems

Empirical Calibration of Ingest and Sensor Risk Coordinates

The foundational pillar of this research initiative is the transformation of sensor and data ingestion calibration from a heuristic, often implicit process into a rigorously defined, quantifiable, and empirically grounded science. The central objective is to create two new, first-class RiskCoordinates within the Cyboquatic ecosystem: `rcalib`, which quantifies ingest-related errors, and `rsigma`, which quantifies sensor uncertainty . These coordinates are not merely diagnostic metrics; they are engineered to serve as direct inputs to the system's primary stability mechanism, the Lyapunov residual, thereby making data trustworthiness a systemic property rather than a local assumption. The process begins with empirical calibration against real-world Phoenix telemetry, moving theoretical concepts into concrete, versioned artifacts that govern the behavior of all industrial nodes. This approach directly addresses the critical challenge of latent data corruption, where undetected issues in data ingestion or sensor drift can silently compromise analysis and lead to unsafe operational states.

The methodology for this empirical calibration is rooted in regression analysis and statistical fitting against historical Phoenix telemetry logs. For the ingest error coordinate, `rcalib`, the process starts by defining an intensity scalar, I_{rcalib}, which is a non-negative weighted sum of various diagnostic counts observed during CSV or `.aln` shard parsing . These counts include fundamental violations such as missing mandatory fields, RFC-4180 specification breaches, type mismatches between expected and actual data, schema inconsistencies, unknown variable IDs, unit mismatches, and the absence of corresponding corridor definitions for a given metric . Each of these error types is assigned a weight $(w_{\text{miss}}, w_{\text{rfc}}, w_{\text{type}}, \ldots)$, which represents its relative severity in the context of the Phoenix basin's operational environment . The empirical step involves analyzing vast quantities of ingest logs from Phoenix nodes to determine the optimal

values for these weights. This is achieved by regressing the frequency and impact of each error type against known failure modes and system performance degradation events. The goal is to construct a kernel where the resulting I_{rcalib} value reflects a meaningful measure of ingest health, allowing for differentiation between benign anomalies and critical data integrity failures.

Once the intensity scalar I_{rcalib} is calculated, it must be normalized into the canonical [0, 1] RiskCoordinate range to be compatible with the existing ecosafety spine . This normalization is performed using a piecewise-affine corridor kernel that maps the raw intensity to the final `r_calib` value based on predefined thresholds: a `safe` threshold, a `gold` threshold, and a `hard` threshold . The empirical calibration phase also involves determining the optimal placement of these three bands. This is done by correlating the distribution of I_{rcalib} values from the telemetry data with system outcomes. The `safe` band would encompass the vast majority of normal operations, the `gold` band would represent conditions that warrant attention but not immediate action, and the `hard` band would correspond to states where the data quality is so poor that it invalidates any downstream analysis. By fitting these bands empirically, the system moves away from arbitrary heuristics to a data-driven definition of what constitutes acceptable versus unacceptable ingest quality. The final output of this process is a versioned artifact, such as `qpudatashards/IngestRcalibPhoenix2026v1.aln`, which freezes these specific weights and corridor bands into a cryptographically signed, replayable shard .

A parallel, though distinct, empirical calibration pipeline is required for the sensor uncertainty coordinate, `rsigma`. This coordinate is designed to capture the combined effect of several physical phenomena that degrade sensor fidelity over time: drift, noise, bias, and loss . The process begins by establishing separate, per-sensor-family corridors for each of these components. For example, flow sensors, pressure sensors, and turbidity sensors will have different characteristics and failure modes, necessitating unique calibration curves for their drift and noise profiles. The empirical fitting for `rsigma` involves a multi-stage process that combines simulation, laboratory testing, and field data collection across diverse basins . Simulation provides a controlled environment to understand the relationship between raw sensor output and the underlying physical error components. Laboratory tests under controlled environmental conditions (e.g., temperature, humidity, vibration) allow for the systematic introduction of known faults to characterize sensor behavior. Finally, long-term field deployments provide the most critical data, capturing the complex interactions of aging, environmental stressors, and operational wear that cannot be fully replicated in a lab.

For each sensor family, the raw measurements of drift, noise, bias, and loss are collected and normalized into their own unitless RiskCoordinates, $r_{\text{drift}}, r_{\text{noise}}, r_{\text{bias}}, r_{\text{loss}} \in [0,1]$.

These individual uncertainties are then aggregated into the composite `rsigma` score using a weighted L2 norm, as specified by the formula:

$$r_\sigma = \sqrt{w_{\text{drift}} r_{\text{drift}}^2 + w_{\text{noise}} r_{\text{noise}}^2 + w_{\text{bias}} r_{\text{bias}}^2 + w_{\text{loss}} r_{\text{loss}}^2}$$

This aggregation method is chosen because it produces a smooth, convex, and 1-Lipschitz function, which is mathematically well-behaved and compatible with the quadratic nature of the Lyapunov residual . The weights $(w_{\text{drift}}, w_{\text{noise}}, w_{\text{bias}}, w_{\text{loss}})$ are again determined empirically through regression analysis against Phoenix and other basin data, reflecting the relative importance of each uncertainty source for a given sensor type. The final `rsigma` value is clamped to 1.0 to maintain it within the canonical RiskCoordinate range . Similar to `rcalib`, the results of this extensive calibration effort are frozen into versioned artifacts, such as `SensorUncertaintyKernel2026v1.aln`, which define the component corridors and weights for every supported sensor family . This ensures that `rsigma` is not just a conceptual tool but a governed, standardized metric that can be computed consistently across the entire Cyboquatic fleet.

The table below outlines the key components and artifacts resulting from this empirical calibration process, providing a clear summary of the transition from abstract theory to concrete, governing rules.

Component	Description	Empirical Fitting Method	Governing Artifact
Ingest Error Kernel (I_{rcalib})	A non-negative weighted sum of diagnostic counts from data ingestion (e.g., missing fields, RFC-4180 violations).	Regression analysis of Phoenix ingest logs correlated with system performance.	Weights (w_k) in `IngestRcalibPhoenix2026v1.aln` .
Ingest Risk Coordinate (r_{calib})	The normalized [0,1] RiskCoordinate derived from the ingest error intensity I_{rcalib}.	Placement of `safe`, `gold`, and `hard` corridor bands based on telemetry distribution and failure correlation.	Corridor bands ($I_{\text{safe}}, I_{\text{gold}}, I_{\text{hard}}$) in `IngestRcalibPhoenix2026v1.aln` .
Sensor Uncertainty Components	Normalized RiskCoordinates for sensor degradation factors: drift, noise, bias, and loss.	Simulation, lab testing, and long-term field data collection to establish per-sensor-family corridors.	Per-component corridor tables in `SensorUncertaintyKernel2026v1.aln` .
Composite Uncertainty Score (r_σ)	The aggregate uncertainty score, calculated as a weighted L2 norm of the component risks.	Regression analysis to determine component weights ($w_{\text{drift}}, w_{\text{noise}}, \ldots$) based on their impact on system reliability.	Composite kernel formula and weights in `SensorUncertaintyKernel2026v1.aln` .

This disciplined, empirical approach fundamentally changes the role of the calibrator. It is no longer a simple utility but the source of truth for data trustworthiness. The resulting

versioned artifacts—such as `IngestRcalibPhoenix2026v1.aln` and `SensorUncertaintyKernel2026v1.aln`—become integral parts of the ecosafety spine, enforceable through the same CI checks that protect the rest of the system . They embody the principle of "no corridor, no build," meaning any deviation from these empirically validated standards requires a formal change proposal that must pass rigorous validation . This grounding in real-world data minimizes subjective assumptions and surfaces the largest sources of residual risk—the numeric placement of corridors and the calibration of weights—directly into the observable state of the system as `rcalib` and `rsigma` values . The knowledge factor for this slice is estimated at 0.95, as it relies heavily on reusing the established grammar and proofs of the existing system while applying them to concrete, newly-calibrated parameters .

Architectural Implementation via Non-Actuating Rust Workloads and ALN Shards

The architectural strategy for implementing the improved calibrator is meticulously designed to align with the core tenets of the Cyboquatic safety framework: immutability, verifiability, and isolation. The chosen substrate is a combination of the Rust programming language and Articulated Language Notation (ALN) shards, deployed as part of a canonical non-actuating workload pattern . This architecture ensures that the calibration logic is computationally sound, securely isolated from any actuation surfaces, and universally consistent across all industrial nodes. By creating a pure Rust crate that operates exclusively on governed data artifacts, the system guarantees that calibration outputs are trustworthy without expanding the hardware's harmful surface area . This design choice is paramount for industrial applications where software defects or unintended behaviors can have significant economic and environmental consequences.

The cornerstone of this implementation is the non-actuating workload pattern, a proven anti-pattern for safely extending the capabilities of industrial machinery . A non-actuating workload is a software module whose sole purpose is to read input data shards, perform computations, and write new output shards. Critically, it contains no logic that interacts with actuators, fieldbus stacks, or any other hardware control interface . The machine-sensor calibrator is positioned as another instance of this pattern. It reads raw telemetry shards (e.g., containing CSV-formatted sensor readings) and/or intermediate diagnostic shards. It parses these inputs using robust Rust libraries compliant with RFC-4180 specifications [37]. Using the empirically calibrated kernels from the frozen `ecosafetycore` library, it computes the `rcalib` and `rsigma` coordinates . The results

—diagnostic counts, `Ircalib`, `rcalib`, `rsigma`, and updated trust-adjusted K, E, and Vt values—are then written back into new, signed shards, tagged with a lane (e.g., RESEARCH, EXP, PROD) and an evidence hash for cryptographic provenance . This entire workflow is executed by a pure Rust crate that has no dependencies on any actuator-related crates, thus satisfying the non-actuation constraint by design .

Rust is selected for this implementation due to its unique combination of memory safety, high performance, and strong static typing. Memory safety, guaranteed at compile time by the ownership and borrowing system, prevents entire classes of vulnerabilities like buffer overflows and use-after-free errors that could corrupt data or destabilize the system . This is crucial when processing potentially malformed or maliciously crafted telemetry. Performance is essential for real-time or near-real-time calibration on resource-constrained industrial nodes. Rust's ability to produce efficient native code makes it suitable for running these computational kernels directly on edge devices. The strong type system complements the schema enforcement provided by ALN. When the Rust crate parses an ALN-defined shard, the compiler can verify that all required fields (like `r_calib` and `rsigma`) are present and correctly typed, catching many potential integration errors before runtime . Furthermore, the crate's dependency graph is strictly controlled; it depends only on the frozen `ecosafety_core` crate, which itself defines the `RiskVector`, `Residual`, and `KerWindow` types, along with the corridor structures . This prevents divergent re-implementations of the risk calculation logic and ensures that every node in the fleet is operating on the exact same monotone, Lyapunov-compatible grammar .

Articulated Language Notation (ALN) serves as the universal, human-readable, and machine-parsable contract that binds the mathematical model to its implementation . The schema for the calibration shards, such as `IngestCalibrationWindow`, is defined in ALN. This `struct` declaration specifies every field, its data type, and its metadata, including whether it is mandatory . For instance, the `IngestCalibrationWindow` struct explicitly includes mandatory fields for `r_calib` and `r_sigma`, ensuring that any shard produced by a compliant calibrator workload must contain these critical data-quality coordinates . The ALN file also contains the kernel definitions themselves. The `IngestErrorKernel2026v1` kernel, for example, contains the explicit formulas for calculating `I_rcalib` from the diagnostic counts and the piecewise-linear formula for mapping `I_rcalib` to `r_calib` using the corridor bands . By encoding the logic directly into the shard's metadata, the ALN file becomes a self-documenting artifact that can be independently verified and replayed. The governance layer, enforced by CI, uses these ALN definitions to check that any new shard being written adheres to the correct schema and contains the necessary coordinates . This creates a closed loop where the

rules are defined in immutable artifacts, implemented in code that respects those artifacts, and checked by automated systems that enforce compliance.

The benefits of this Rust and ALN-based substrate are multifaceted, directly addressing key challenges in industrial IoT and safety-critical systems.

- **Pre-processing Ingest Quality:** The `rcalib` coordinate acts as a hard brake on data ingestion quality. Any violation of RFC-4180 or schema fidelity is propagated directly into a visible `[0, 1]` risk coordinate, preventing bad data from corrupting subsequent analysis . The eco-impact is high (0.92) as it turns bad data into an immediately actionable signal across the fleet .
- **Sensor Uncertainty Quantification:** The `rsigma` score bundles complex sensor degradation effects into a single, convex, and Lipschitz-continuous score. This score is integrated into the Lyapunov residual, ensuring that increasing uncertainty directly impacts the system's perceived stability .
- **Governance and Reproducibility:** Because the kernels, corridor bands, and weights are all bound to DID-signed, replayable ALN shards, the entire calibration process becomes auditable and reproducible . Changes require creating new versioned artifacts, which are subject to the same governance as the rest of the ecosafety spine.
- **Industrial Integration:** The non-actuating nature of the Rust workload allows OEMs to integrate this technology into their proprietary controllers. They can run the same Rust calibrator crate to compute `rcalib` and `rsigma` locally, feeding these trusted data-quality signals into their own decision-making processes without ever needing access to sensitive actuator APIs . This promotes interoperability and standardization.
- **Eco-Risk Traceability:** By baking data-quality coordinates directly into the RiskVector and KER state, any calibration error or data corruption becomes impossible to hide in post-analysis. The full history of data trustworthiness is cryptographically chained to the decisions made based on that data .

This architectural pattern has a high knowledge factor (0.95) as it is a direct mirror of the existing EcoNetSchemaShard2026v1 and ecosafetycore v2 patterns . The residual risk is low (around 0.12) and is explicitly centered on the numeric fit of the corridors, which are now shard-visible and subject to rigorous replay testing . The table below summarizes the gains from this architectural choice.

Aspect	Benefit Summary	Supporting Concepts
Pre-processing Ingest Quality	Propagates any CSV/`.aln` error into a visible `[0, 1]` risk coordinate (`rcalib`).	RFC-4180 compliance, mandatory schema fields, diagnostic count aggregation .
Sensor Uncertainty Quantification	Bundles drift, noise, bias, and loss into a convex, 1-Lipschitz score (`rsigma`) integrated into `Vt`.	Weighted L2 norm, component-wise corridor mapping, empirical fitting .
Governance and Reproducibility	Binds kernels, bands, and weights to DID-signed, replayable ALN artifacts.	"No corridor, no build", shard signing, cryptographic provenance .
Industrial Integration	Allows OEMs to expose calibrator outputs without actuator control.	Non-actuating workload pattern, pure Rust crate, lane tagging .
Eco-Risk Traceability	Makes calibration errors impossible to hide in post-analysis by integrating `rcalib` and `rsigma` into the core state.	Cryptographic chaining, lane-tagged shards, evidence hexes .

By implementing the calibrator as a non-actuating Rust workload governed by ALN shards, the project achieves a powerful synthesis of modern software engineering practices and classical control theory principles. This foundation is essential for the next critical step: wiring these new data-quality coordinates into the system's primary stability and deployment mechanisms.

Integrating Data Quality into the Lyapunov Residual and KER Gates

The most profound contribution of this research lies in its strategic integration of the newly defined data-quality coordinates, `rcalib` and `rsigma`, into the core stability and deployment logic of the Cyboquatic system. This is not a superficial addition but a deep architectural change that elevates data trustworthiness from a secondary concern to a primary determinant of the system's state and operational envelope. By formally incorporating `rcalib` and `rsigma` into the Lyapunov residual (`Vt`) and the KER (Keep, Err, Risk) deployment gates, the system enforces a strict "only down-credit, never inflate" behavioral guarantee . This means that any degradation in data quality can only make the system appear less safe or keep its current state; it can never cause the system to be perceived as more reliable or permit more aggressive actions. This section details the mathematical formulation of this integration and explains how it operationalizes the principle of verifiable safety.

Mathematically, the system's stability is governed by the discrete-time Lyapunov residual, $V_t = \sum_j w_j r_{j,t}^2$, where r_j are the normalized RiskCoordinates for various physical planes (hydraulics, biology, etc.), and w_j are their corresponding positive Lyapunov weights .

The `safestep` invariant mandates that, outside of a small stable interior region, the residual must be non-increasing: $V_{t+1} \leq V_t$. The proposed enhancement extends this residual to a full-data residual, V_t^{full}, by adding dedicated terms for the data-quality planes:

$$V_t^{\text{full}} = \sum_{j \text{ phys}} w_j r_{j,t}^2 + w_{\text{calib}} r_{\text{calib},t}^2 + w_\sigma r_{\sigma,t}^2,$$

where w_{\text{calib}} > 0 and w_\sigma > 0 are the positive weights for the `rcalib` and `rsigma` coordinates, respectively . This extension is provably safe and preserves the desirable properties of the original residual. Since the weights are positive and the coordinates are bounded in `[0, 1]`, the added quadratic terms are always non-negative, ensuring that V_t^{full} remains non-negative and convex . Crucially, because the `safestep` logic already operates on the extended state space, the invariant $V_{t+1}^{\text{full}} \leq V_t^{\text{full}}$ continues to hold. This creates a powerful feedback loop: if data quality degrades (i.e., `rcalib` or `rsigma` increases), the full residual `Vt_full` must increase or remain constant. According to the `safestep` rule, this forces the system to derate or stop, preventing the propagation of unreliable information into the control loop . This transforms the Lyapunov function from a measure of physical stability alone into a holistic measure of both physical and informational stability.

The integration with the KER deployment gates is equally critical for enforcing the "only down-credit, never inflate" policy. The KER state, maintained by a sliding window over recent timesteps, consists of three primary metrics: `K` (the fraction of steps passing `safestep`), `R` (the maximum RiskCoordinate observed in the window), and `E` (the ecological metric, defined as $E = 1 - R$) . The window length is denoted as N. To incorporate data quality, two key modifications are made. First, the calculation of `R` is expanded to include all relevant coordinates, including `rcalib` and `rsigma`:

$$R = \max_{j,t} r_{j,t},$$

where the maximum is taken over all physical and data-quality planes (`j` in {phys, calib, sigma}) and all timesteps `t` in the window . This ensures that a spike in ingest errors or sensor uncertainty is immediately reflected in the `R` metric, signaling a broader loss of confidence in the system's state estimate.

Second, and more subtly, the raw `K` and `E` scores are adjusted using a data trust scalar. The data trust scalar, D_{data}, is defined as $1 - r_{\text{calib}}$. This scalar represents the degree of

trust in the integrity of the ingested data, ranging from 1.0 (perfect ingest quality) to 0.0 (completely untrustworthy data). The adjusted K and E scores are then calculated as:

$$K_{\text{adj}} = K_{\text{raw}} D_{\text{combined}}, \quad E_{\text{adj}} = E_{\text{raw}} D_{\text{combined}},$$

where D_{combined} is an overall trust product that may also include a sensor trust component, D_{sensor} . This adjustment is the mathematical embodiment of the "only down-credit" principle. Since D_{data} is a value in `[0, 1]`, multiplying `K_raw` and `E_raw` by it can only decrease or preserve their values; it can never increase them . Therefore, if `rcalib` increases (worsening data quality), D_{data} decreases, which causes K_{adj} and E_{adj} to decrease or stay the same. Simultaneously, as established above, the increase in `rcalib` will weakly increase `R` (and thus weakly decrease `E = 1-R`). The net effect is that any data-quality issue manifests as a degradation in multiple, independent metrics: `R` goes up, and `K_adj` and `E_adj` go down. No amount of favorable physical-plane data can mask a decline in data trust.

This integrated system of `Vt_full` and trust-adjusted KER gates establishes a robust hierarchy of safety. The `kerdeployable` invariant, which gates production lane activation, is extended to explicitly check these data-quality coordinates. For a deployment decision to be approved in a PROD lane, the calibrator's output must satisfy stringent conditions. The `kerdeployable` logic must enforce explicit hard caps, such as requiring `r_calib` to be well inside its "gold" band and `rsigma` to be below its "hard" band before any actuators are enabled . This makes calibration quality a hard gate, entirely independent of how well the physical risk planes look. An industrial machine could appear physically stable but would be prevented from operating in a production capacity if its calibrator detects poor data ingest or excessive sensor uncertainty. This prevents the silent accumulation of risk that can occur when relying solely on physical sensors.

The table below summarizes the mathematical relationships and behavioral guarantees of this integrated system.

System Component	Mathematical Formulation	Behavioral Guarantee / Purpose
Full Lyapunov Residual	$V_t^{\text{full}} = \sum_{j\ \text{phys}} w_j r_{j,t}^2 + w_{\text{calib}} r_{\text{calib},t}^2 + w_\sigma r_{\sigma,t}^2$	Ensures worse data quality (`↑rcalib`, `↑rsigma`) weakly increases `Vt`, forcing `derate/stop` via `safestep`.
KER Window Max (R)	$R = \max_{j,t} r_{j,t}$	Ensures R captures the worst-case scenario across all planes, including data quality (`rcalib`, `rsigma`).
Data Trust Scalar (`D_data`)	$D_{\text{data}} = 1 - r_{\text{calib}}$	Quantifies trust in ingested data, acting as a multiplicative down-credit factor.
Trust-Adjusted K/E (`K_adj`, `E_adj`)	$K_{\text{adj}} = K_{\text{raw}} D_{\text{combined}}$, $E_{\text{adj}} = E_{\text{raw}} D_{\text{combined}}$	Guarantees that worsening data quality can only reduce K and E; it can never improve them ("only down-credit").
Deployment Gate (`kerdeployable`)	Requires `r_calib <= r_calib_hard_max` and `r_sigma <= r_sigma_hard_max` for PROD lanes.	Enforces data quality as a hard, independent gate for production deployment, regardless of physical plane status.

The mathematical proofs supporting this integration are largely direct reuses of the existing system's grammar and proofs, giving it a high knowledge factor (approx. 0.96) . The eco-impact is also very high (0.93) because it makes calibration quality a first-class brake on deployment, directly reducing the risk of deploying on faulty data . The residual risk is low and well-contained (0.11–0.13), as it is continuously and visibly represented by the coordinates `rcalib`, `rsigma`, `Vt`, and R, which are the very metrics being monitored . This complete integration ensures that the system's response to data quality issues is not just reactive but is woven into the very fabric of its stability and safety logic.

Governance and Validation through Monotonic Fault-Injection Replays

To move beyond theoretical designs and achieve verifiable safety, the proposed calibration framework must be subjected to rigorous, automated validation. The core of this validation strategy is the creation of universal, versioned fault-injection replay harnesses. These harnesses serve as the ultimate proof that the system behaves monotonically, adhering to the "only down-credit, never inflate" guarantee . By replaying historical Phoenix telemetry—which encompasses a wide variety of operational and fault conditions—and injecting synthetic errors, these harnesses test the system's response under controlled scenarios. Any behavior that violates the expected monotonicity—increase in risk coordinates, decrease in trust metrics—is flagged as a regression, failing the continuous integration (CI) pipeline. This approach operationalizes the mathematical

proofs of safety, turning them into executable, repeatable tests that can be applied across the entire Cyboquatic ecosystem.

The design of these replay harnesses is systematic and comprehensive. The foundation is a set of canonical datasets, primarily composed of historical telemetry from Phoenix nodes, stored as governed `.aln` shards . These shards contain not only the sensor readings (`rx`) but also the associated Lyapunov residual (`Vt`), KER state, and evidence hashes, ensuring a complete and verifiable snapshot of past system states . The replay harness takes one of these historical windows as its baseline. It then proceeds to inject a library of pre-defined synthetic faults into the data stream. These fault models are representative of common issues in industrial IoT systems and are categorized into ingest-level and sensor-level faults [9] .

For ingest-level faults, the harness simulates problems in the data acquisition and parsing stages. These include:

- **Schema Violations:** Intentionally omitting mandatory fields from a shard or altering the data type of a field to trigger a type mismatch error .
- **RFC-4180 Compliance Failures:** Inserting lines that violate the comma-separated values standard, such as unquoted newlines or improper quoting, to test the robustness of the CSV parser [37] .
- **Metadata Corruption:** Changing the `varid` of a field or removing its corresponding entry from the corridor database to simulate a misconfigured sensor .

For sensor-level faults, the harness modifies the raw sensor data to emulate physical degradation and environmental interference. These simulations are informed by the empirical calibration of the `rsigma` components . Faults include:

- **Simulated Drift:** Gradually shifting the mean value of a sensor reading over time.
- **Noise Injection:** Adding random fluctuations of varying amplitude to a clean signal.
- **Bias Introduction:** Applying a constant offset to a sensor's output.
- **Signal Loss:** Periodically setting a sensor's value to null or an extreme outlier.

After injecting a fault, the harness re-runs the non-actuating calibrator workload on the modified dataset. The output is compared against the original, unmodified replay to assert a series of critical invariants. These assertions are the heart of the validation process, providing quantitative proof of the system's desired behavior.

The table below details the fault injection scenarios and the corresponding invariants that must be upheld.

Fault Category	Specific Fault Injection Example	Asserted Monotonicity Invariant

Ingest Faults	Missing a mandatory field in the input shard.	`rcalib_new >= rcalib_old`
		`Vt_full_new >= Vt_full_old`
		`R_maxcoord_new >= R_maxcoord_old`
		`K_adj_new <= K_adj_old`
		`E_adj_new <= E_adj_old`

Ingest Faults	Injecting an RFC-4180 violation.	Same as above.

Sensor Faults	Injecting moderate noise into a pressure sensor signal.	`rsigma_new >= rsigma_old`
		`Vt_full_new >= Vt_full_old`
		`R_maxcoord_new >= R_maxcoord_old`
		`K_adj_new <= K_adj_old`
		`E_adj_new <= E_adj_old`

Sensor Faults	Simulating a 10% sensor drift.	Same as above.

Corridor Evolution	Tightening the `rcalib` 'gold' band in the governing shard.	Replay with old bands vs. new bands must show:
		`K_adj_new >= K_adj_old`
		`E_adj_new >= E_adj_old`
		`R_maxcoord_new <= R_maxcoord_old`

The first four rows of the table represent the core "only down-credit" guarantee. The harness proves that introducing a fault weakly increases all risk metrics (`rcalib`, `rsigma`, `Vt_full`, `R`) and weakly decreases all trust metrics (`K_adj`, `E_adj`). If any assertion fails—for example, if `K_adj` were to increase after injecting a sensor fault—the CI job fails, and the change that introduced the faulty corridor or kernel is rejected. This prevents any regressions that would cause the system to become less safe while appearing more reliable.

The fifth row addresses a more subtle but equally important aspect of governance: the evolution of corridors themselves. The "only tighten corridors" invariant is a cornerstone of the ecosafety spine . The replay harness is used to validate this principle. When a developer proposes tightening a corridor (e.g., making the `rcalib` 'gold' band smaller), the harness replays a fixed window of Phoenix data using both the old and new versions of the governing corridor shard. The assertion is that tightening the corridor should only make the system look safer or leave it unchanged. Specifically, the adjusted K and E metrics should weakly increase (or stay the same), and the adjusted R metric should weakly decrease (or stay the same). If tightening a corridor causes `K_adj` to drop or `R` to rise for a given set of physical evidence, it indicates a problem with the corridor placement, and the change is blocked . This provides a powerful, data-driven defense against poorly calibrated safety margins.

These universal replay harnesses are designed to be versioned CI actions that are shared across all ecosafety-aware repositories . This ensures consistency and eliminates the possibility of different teams validating the same logic in different ways. The existence of these harnesses provides a high degree of confidence in the system's safety properties. The knowledge factor for this slice is around 0.95, as it builds upon the existing Phoenix pilot and non-actuating workload patterns . The eco-impact is substantial (0.92), as it reduces long-term harm risk on industrial hardware by systematically identifying and blocking dangerous regressions . The residual risk is low (0.12) and is explicitly centered on the calibration of new sensor types and the tuning of corridors, which are precisely the areas where these harnesses provide their greatest value by surfacing issues early . By combining empirical data, formal mathematical guarantees, and automated, fault-injection-based verification, this governance and validation layer provides a robust, multi-faceted defense for the entire system.

Industrial Deployment on Cyboquatic Nodes and Strategic Roadmap

The successful implementation of the enhanced calibrator framework culminates in its deployment across the entire portfolio of industrial-grade Cyboquatic nodes, including FlowVacs, MAR nodes, and canal retrofits. This final stage translates the theoretical constructs and validated CI pipelines into tangible improvements in the precision and safety of real-world industrial machinery. The deployment strategy leverages the existing non-actuating workload pattern, ensuring that the calibrator integrates seamlessly into the operational telemetry pipeline of each node without introducing any new actuation surfaces or expanding the harmful hardware-software interface . The rollout is managed through a controlled, governed path that allows for the gradual sharpening of calibration over time, anchored firmly to the Phoenix data corpus and validated by the universal replay harnesses.

On each industrial node, the machine-sensor calibrator functions as a dedicated, non-actuating workload attached to the main telemetry processing pipeline . At the ingest stage, the calibrator workload reads the raw CSV or `.aln` telemetry streams. It immediately begins computing diagnostic counts for any schema or RFC-4180 violations and flags them directly within the shard being processed . This ensures that every data source, whether it is a primary sensor or a legacy system interface, propagates its calibration risk into the node's global `RiskVector`. On the sensor side, the workload ingests calibration histories and statistics from sliding windows of telemetry. It then

applies the empirically calibrated `rsigma` composite kernel to bundle the normalized drift, noise, bias, and loss components into a single `rsigma` score . The resulting `rcalib` and `rsigma` values, along with the trust-adjusted K, E, and Vt, are written into governed shard families (e.g., `qpudatashards/IngestTrustPhoenix2026v1.aln`) that feed into the central `KerWindow` and `kerdeployable` logic . Throughout this process, the OEM controllers and other subsystems see only the finalized, trust-adjusted state of the `RiskVector` and KER, with the data-quality calculations having already been completed and baked in. The harmful surface remains frozen behind the `safestep` and `kerdeployable` walls, which now have the additional, powerful brakes provided by `rcalib` and `rsigma` .

The controlled path for sharpening calibration over time is a critical feature of this deployment strategy. Because all calibrator components—including the Rust crate, the ALN corridor shards, and the replay harnesses—are governed by the same CI system that protects the ecosafety spine, any improvement must be rigorously tested before it can be merged into the main branch . For example, if a team identifies an opportunity to improve the `rsigma` kernel for a specific sensor family, they must follow a clear process. They would first run the sensor through the full simulation-to-lab-to-field calibration pipeline to derive new component corridors and weights . They would then update the `SensorUncertaintyKernel2026v1.aln` shard with these new values. Before the change can be merged, it must pass a CI job that runs the universal fault-injection replay harnesses over a large corpus of historical Phoenix data. The harnesses will verify that the new kernel maintains monotonicity: that worse data still leads to higher risk and lower trust, and that tightening the new corridors does not make bad data look safer . Only after passing these automated, data-driven proofs is the change considered safe to propagate to the entire fleet. This turns the improvement process from an informal exercise of "researching it" into a formal, leveraged, and auditable procedure .

Based on the analysis of the provided materials and the prioritization criteria, the strategic roadmap for completing this research and achieving full implementation can be summarized in the following highest-yield objectives:

1. **Finalize Empirical Calibration Artifacts:** The immediate priority is to complete the empirical fitting of the `rcalib` and `rsigma` kernel weights and corridor bands using Phoenix ingest logs and sensor data . This involves running the necessary regression analyses and calibration pipelines to freeze the definitive parameters for the Phoenix basin. The deliverables are the versioned ALN particles, such as `IngestRcalibPhoenix2026v1.aln` and `SensorUncertaintyKernel2026v1.aln`, which will serve as the canonical reference for all future development. This step removes the largest remaining

source of unconstrained variables and provides concrete parameters for the subsequent proofing steps .

2. **Formalize Mathematical Proofs and ALN Invariants:** Once the empirical parameters are frozen, the next step is to lift the existing mathematical arguments into formal, named lemmas with unique identifiers (hex IDs) . Key lemmas to be created include: (1) a proof that the extended Lyapunov residual V_t^{full} remains non-increasing under `safestep`; (2) a proof of monotonicity for the `rcalib` and `rsigma` kernel mappings; and (3) a proof of the "only down-credit" property for the trust-adjusted K and E metrics. These lemmas, along with their associated mathematical formulas, must be documented in the `ecosafety-core` documentation and tied to the specific Rust functions and ALN invariants they describe . This step transforms the design from a well-reasoned concept into a formally verified component of the system.

3. **Implement Universal Fault-Injection Replay Harnesses:** The third priority is to build and integrate the universal fault-injection replay harnesses into the CI system . This involves writing the CI jobs that orchestrate the replay of historical Phoenix data, inject the library of synthetic faults (both ingest and sensor-based), and execute the assertions on the resulting risk coordinates . These harnesses are the operational backbone of the "only down-credit" guarantee. Their implementation provides the empirical, data-driven validation that the system behaves as intended under fault conditions, serving as the ultimate defense against silent regressions .

4. **Propagate Mandatory Fields and Update Deployment Gates:** Concurrently, the `rcalib` and `rsigma` coordinates must be propagated across all relevant shard families in the Cyboquatic ecosystem. This requires updating numerous shard schemas (e.g., in `EcoNetSchemaShard2026v1` and `ecosafety.corridors.v2`) to mark these fields as mandatory . Correspondingly, the `kerdeployable` invariant must be updated to include explicit, band-based gates for data quality, such as `BlockedByCalib` and `BlockedByDataQuality`, especially for production lanes . This ensures that the entire system, not just the Phoenix pilot, begins to leverage the new data-quality planes.

By executing this roadmap, the project successfully achieves its overarching goal. It transforms machine-sensor calibration from a latent assumption into a transparent, quantifiable, and computationally enforced safety plane. The resulting system is demonstrably safer, as it has a provable mechanism to detect and respond to data quality

degradation. It is more precise, as the reduction of interpretation drift between pre-processing and post-analysis leads to more accurate downstream analytics. And it is more governable, as the entire calibration logic is encapsulated in versioned, auditable artifacts that are protected by a robust, automated validation framework.

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